



EUROPEAN COALITION FOR AI
IN FUNDAMENTAL PHYSICS
CONFERENCE
EuCAIFCon 2026

CNN-Based Triggers for Quark-Gluon Plasma - Sensitive Event Selection in Heavy-Ion Experiments

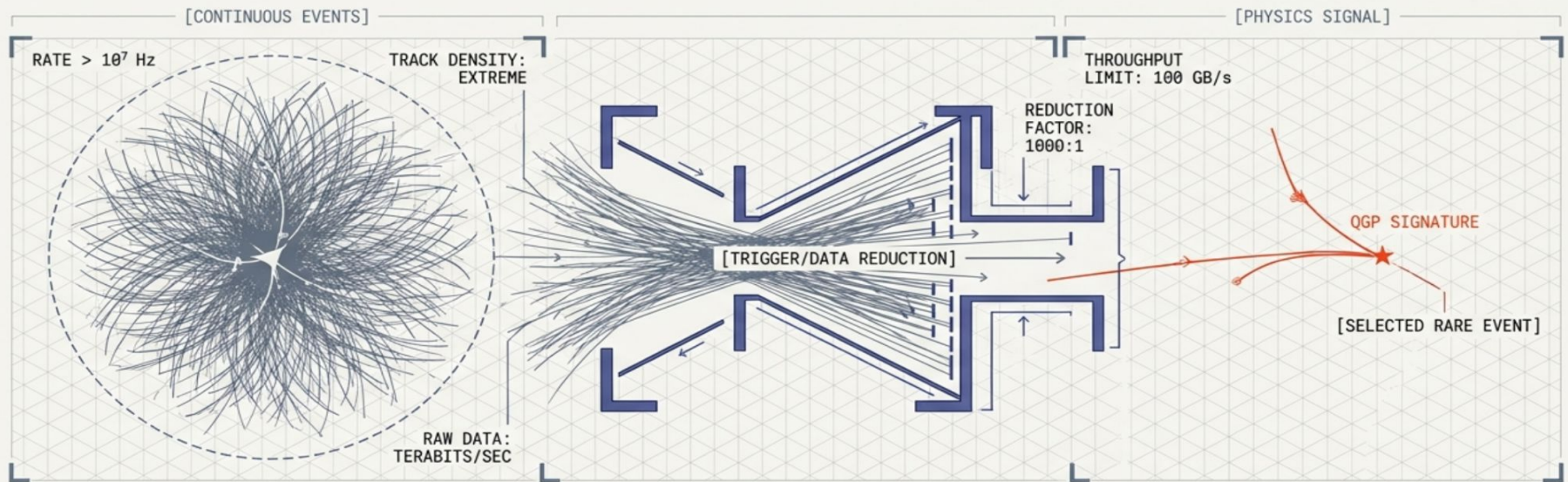
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Based on [2605.26280](#)



26.08.26
Poster 21



The high-rate trigger challenge



[ENVIRONMENT]

Continuous stream of reconstructed events generated in high-rate experimental facilities like CBM at FAIR.



[CONSTRAINT]

Stringent data throughput and storage limits require rapid, online data reduction.

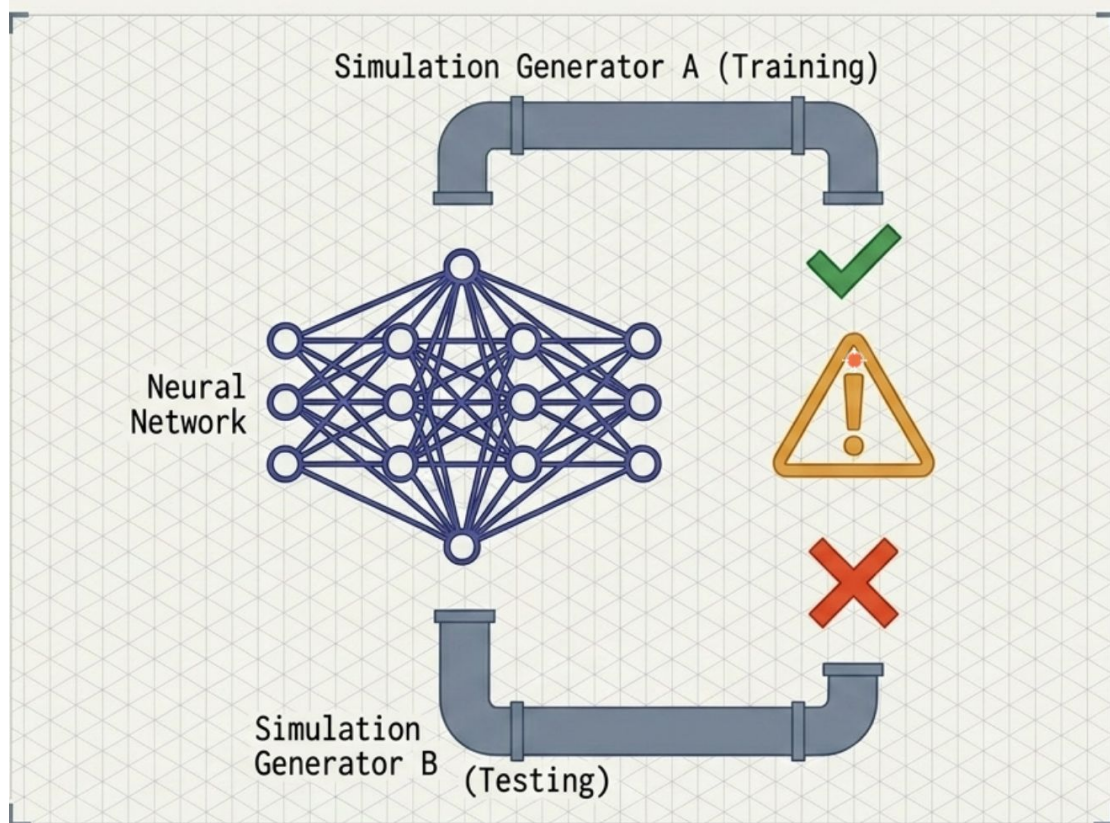


[OBJECTIVE]

Identify rare Quark-Gluon Plasma (QGP) signatures in real-time, software-based analysis chains.



Danger of Model Bias



THE TRAP

Neural Networks often learn model-specific correlations and simulation artifacts rather than true generic signatures of deconfined matter.

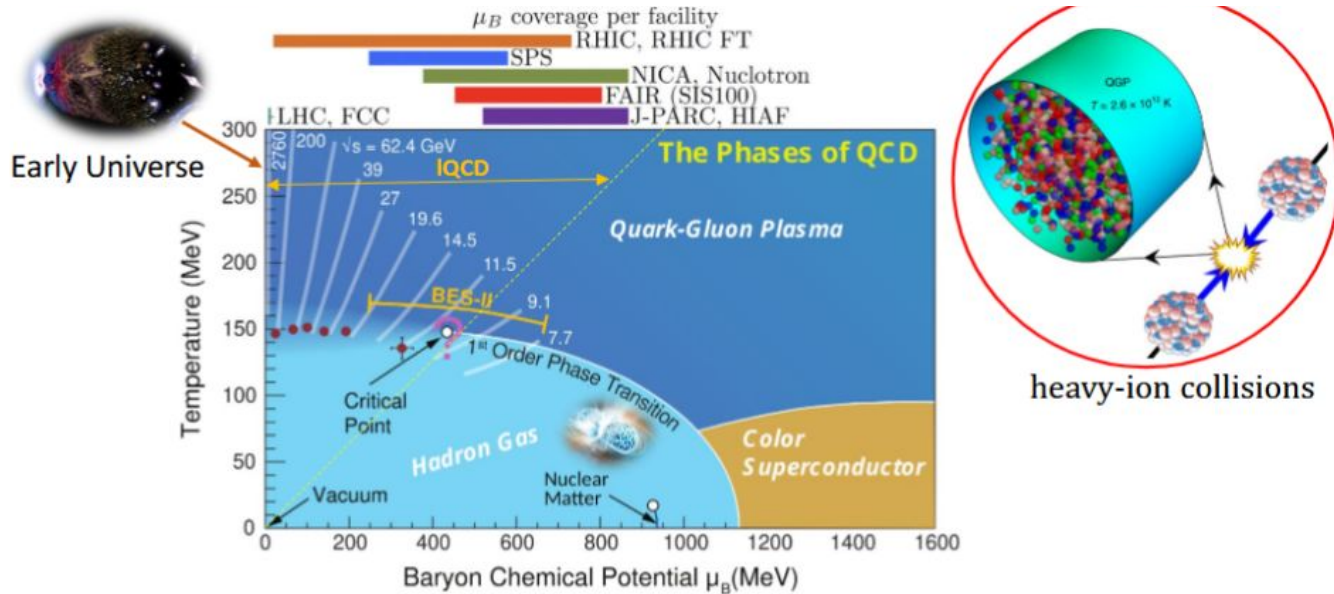
THE REQUIREMENT

Model independence is not an auxiliary property—it is a central requirement for experimental applicability.

THE GOAL

Build a classifier that extracts robust physical features, immune to the underlying physics assumptions of the training data.

Quark-Gluon Plasma Search in HIC Experiments



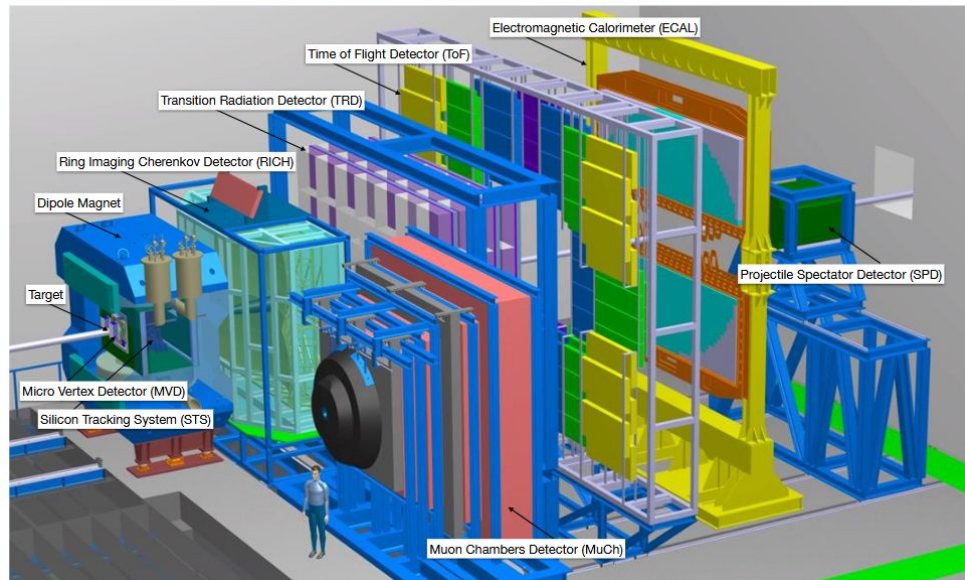
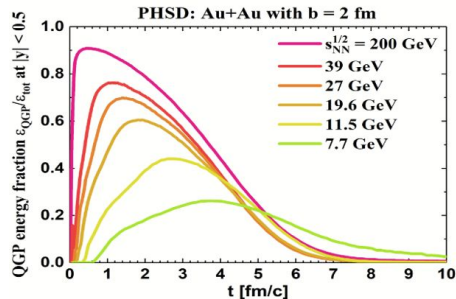
- Explore the QCD Phase Diagram
- Search for CEP and 1st order Phase Transition
- Quantify properties of created hot and dense QGP

CBM experiment at FAIR (GSI)

Main objective – to quantify the properties of created Quark-Gluon Plasma

! Due to the **extremely high rate**, only a small fraction of collisions can be stored for the analysis

Real time data selection and triggering play a crucial role!



For CBM Physics – volume of QGP is essential
QGP fraction at FAIR energies are extremely small
Online QGP-sensitive selection is needed

Multi-model validation

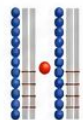


PHSD (Parton-Hadron-String Dynamics)

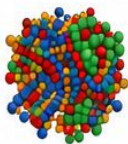


Stages of Collision (PHSD)

Initial A+A Collision
(Strings)



QGP State
(DQPM Partons)



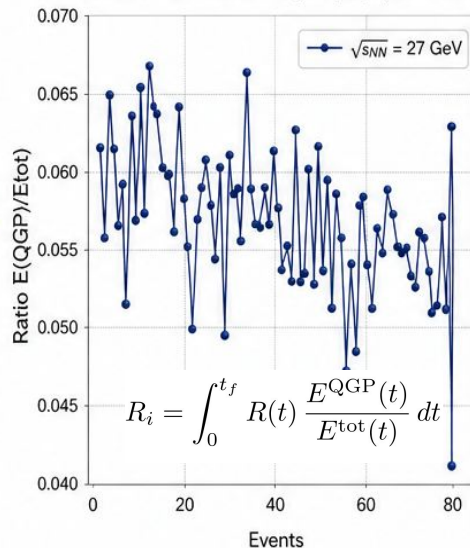
Hadronization
(Clustering)



Hadronic Phase
(HSD)



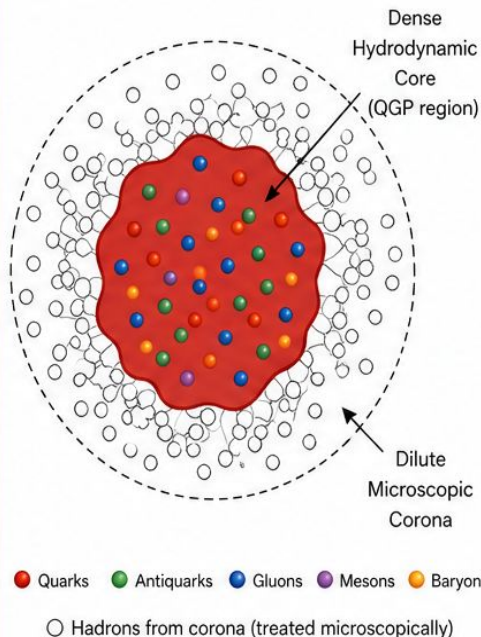
Ratio from hadronisation $E(\text{QGP})/E(\text{tot})$ vs. Events



HSD - W. Cassing and E. L. Bratkovskaya, Nucl. Phys. A 831, 215 (2009);
PHSD 5 - P. Moreau, O. Soloveva, et al, PRC 2019

Microscopic off-shell transport providing **fully dynamical** event-by-event collision histories with **direct QGP labels** based on local energy density.

UrQMD (Ultra-relativistic Quantum Molecular Dynamics)



Key Idea

UrQMD models the dense, early stage of the collision with a hydrodynamic core, while the dilute outer region (corona) is treated microscopically.

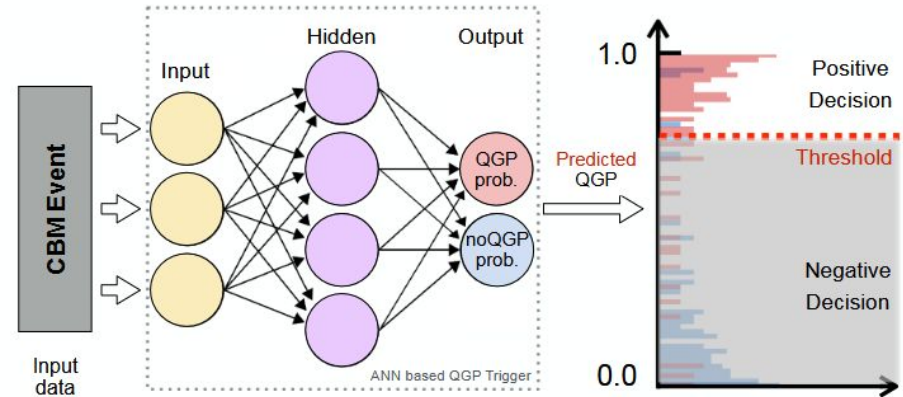
It provides **explicit QGP particle labels (quarks and gluons)** in the hydrodynamic core, enabling partonic degrees of freedom in the early stage.

Hadronic/hybrid transport. The QGP phase is modeled macroscopically in the **hydrodynamic core** (providing **explicit QGP particles: quarks and gluons**), while the dilute **corona** is treated microscopically.

UrQMD / core-corona J. Steinheimer and M. Bleicher, PRC **84**, 024905 (2011), arXiv:1104.3981.
UrQMD with chiral mean-field EoS: M. Omana Kuttan, A. Motornenko, et al, Eur. Phys. J. C **82**, 427 (2022), arXiv:2201.01622

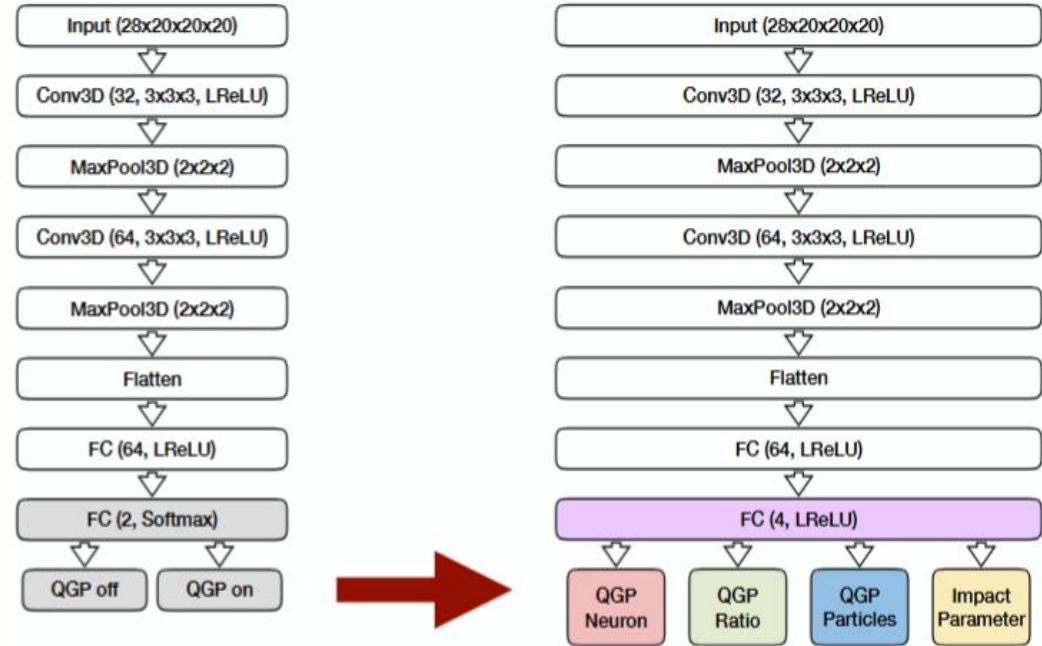
CNN- trigger concept for online QGP event selection in the CBM

- The NN processes reconstructed event data and predicts the probability that an event contains QGP
- A predefined threshold determines - the event is classified as QGP/noQGP
- Only events above the threshold are selected for storage and analysis
- Varying the threshold allows control of trigger sensitivity and selection efficiency



CNN scheme for PHSD

- The CNN model was updated to work with new PHSD data and predict several physical parameters for each event
- The output layer now has 4 neurons that give the QGP score, ratio, number of QGP particles, and the impact parameter.
- The activation function-**LeakyReLU**, which keeps the predicted values positive and physically realistic.
- The loss function was switched to MSE, making the training more stable for continuous outputs



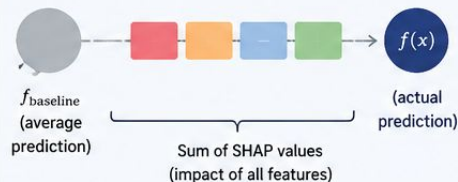
Updated CNN architecture

SHAP (SHapley Additive exPlanations)

A game-theoretic approach to explain machine learning predictions

1. Intuition

SHAP values explain a prediction by measuring how much each feature contributes to moving the prediction from the average (baseline) output to the actual prediction.



2. The Idea (Shapley Value)

Imagine each feature as a player in a coalition game. We want to fairly distribute the “payout” (the prediction) among all players based on their contribution.



Payout = $f(x) - f_{\text{baseline}}$
(Total contribution above baseline)

3. How SHAP Values are Calculated

For each feature i , SHAP value ϕ_i is the average marginal contribution of that feature across all possible coalitions (all possible subsets of features).

Formula:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

Where:

- N = set of all features
- S = a subset of features not containing i
- $f(S)$ = model prediction using features in S
- $f(S \cup \{i\}) - f(S)$ = marginal contribution of i to S
- The fraction is the weighting term from Shapley value theory

Example with 3 Features (A, B, C)

Coalition S (without i)	Add feature i	Marginal Contribution
$\{\}$	$\{i\}$	$f(\{i\}) - f(\{\})$
$\{A\}$	$\{A, i\}$	$f(\{A, i\}) - f(\{A\})$
$\{B\}$	$\{B, i\}$	$f(\{B, i\}) - f(\{B\})$
$\{A, B\}$	$\{A, B, i\}$	$f(\{A, B, i\}) - f(\{A, B\})$

Average (weighted) of above = ϕ_i

4. Final Explanation (Additive)

The prediction is explained as the sum of the baseline value and all SHAP values.

$$f(x) = f_{\text{baseline}} + \sum_{i=1}^M \phi_i$$

Diagram illustrating the final explanation formula. The formula is shown in a green box. Below the formula, arrows point from the terms to their interpretations: f_{baseline} points to 'Average prediction (baseline)', $\sum_{i=1}^M \phi_i$ points to 'Sum of feature contributions (SHAP values)', and $f(x)$ points to 'Actual prediction'.

5. Interpretation

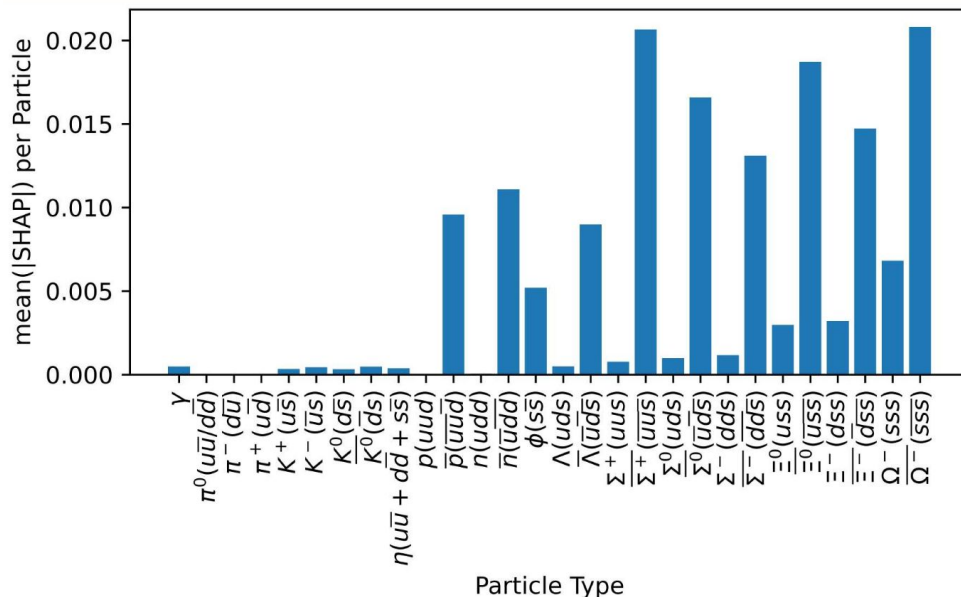
- +** Positive SHAP value ($\phi_i > 0$): feature pushes the prediction higher.
- Negative SHAP value ($\phi_i < 0$): feature pushes the prediction lower.

Visual Example (waterfall plot)



Shapley Additiv Explanations Values(SHAP)

- SHAP method to interpret the NN
- Each input feature acts as a player contribution to the model decision
- SHAP values shows how each feature influences the prediction toward QGP/non QGP classification
- This approach links ML decisions with physical QGP observables



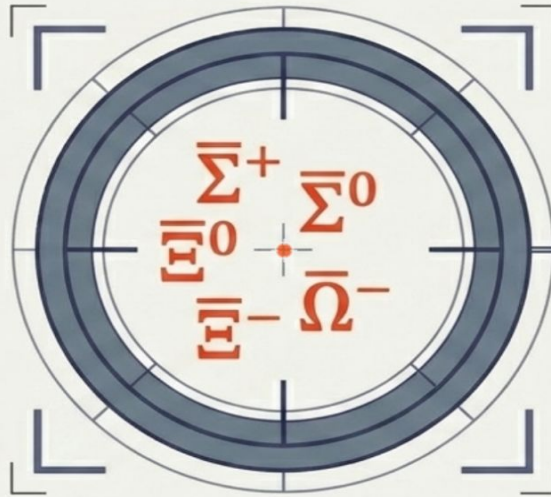
SHAP analysis highlights that strange and antyquark observables contribute the most to QGP prediction

The Physics Fingerprint: Learning Strangeness

THE DOMINANT FEATURES

Multi-strange and strange antibaryons provide the strongest contributions. Light mesons have negligible impact.

Targeting Reticle



THE PHYSICS REALITY

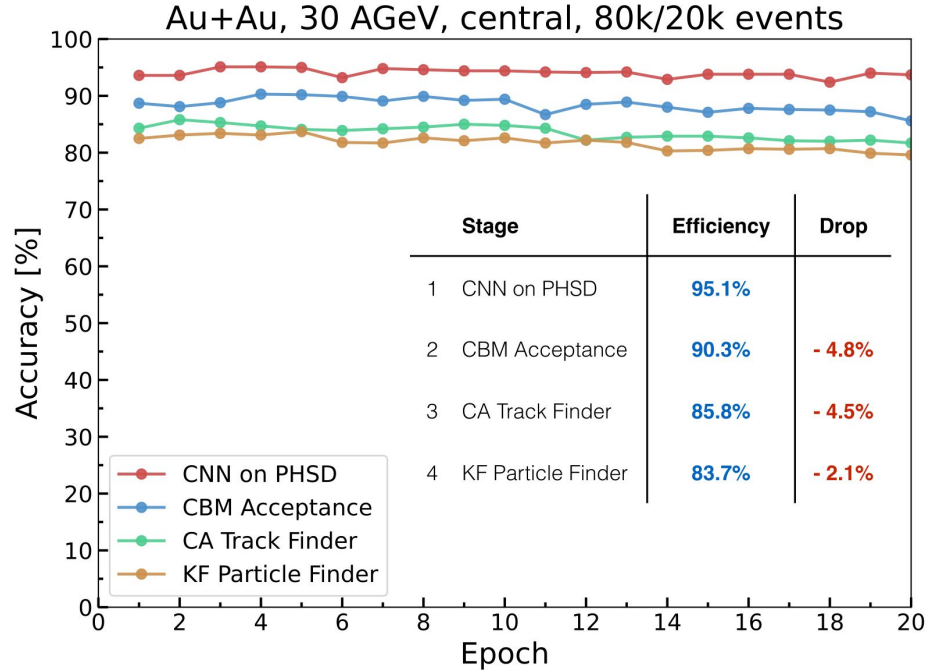
Enhanced strangeness production is a characteristic, well-documented signature of the Quark-Gluon Plasma phase.

[QGP CONFIRMATION YIELD]

THE BREAKTHROUGH

This sensitivity was not programmed. The CNN organically learned that strangeness enhancement is the key signature of QGP purely from multidimensional particle distributions.

Classification accuracy in the CBM Experiment



CNN keeps high resolution even after full reconstruction -> proves robustness to detector effects and suitability for real time QGP event selection

Conclusions and Outlook

- CNN based event selection can identify collisions with QGP from reconstructed final/state particles
- High accuracy is preserved after full reconstruction -> robustness to detector effects and real/time applicability
- PHSD-UrQMD cross/validation demonstrates model independence
- SHAP analysis connects the CNN response to physically meaningful particle features, especially strange hadrons and antibaryons

Work in progress

Testing on GPU

GNN + CNN - similar results

Transformers

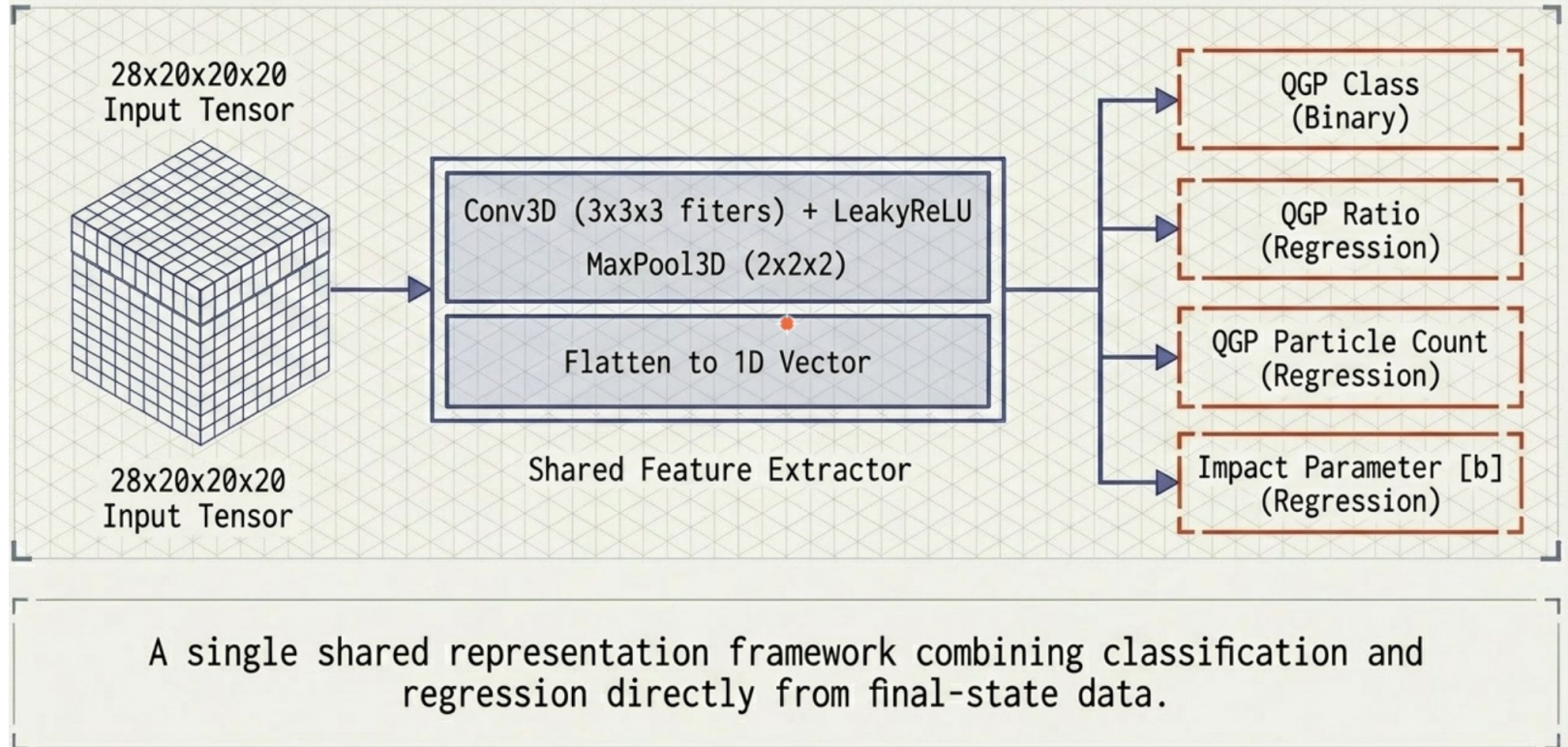
Improvement from Physics/Simulations

Thank you for your attention!

olgasolovev.github.io



Input data



Multi-Model training and validation

